OGOBUCHI DANIEL OKEY

B.Eng. (Hons), MSc. (Brazil), MIEE, MIAENG, MNSE, MNIPES, MACSP, COREN Reg.
Okey.ogobuchi@mouau.edu.ng | ORCID | Google Scholar | CNPq Lattes

EDUCATION

Universidade Federal do ABC, Santo Andre, Sao Paulo, Brazil **2022**
Ph.D. Information Engineering (Thesis presentation, September 2026)
Advisor: Professor João Henrique Kleinschmidt

Universidade Federal de Lavras, Minas Gerais, Brazil **2022**
MSc. Systems Engineering and Automation (Grade: 9.1/10)
Advisor: Professor Demóstenes Zegarra Rodríguez

Michael Okpara University of Agriculture, Umudike, AB, NG **2017**
B.Eng. (Hons) Computer Engineering (CGPA: 4.50/5.0)
Advisor: Professor Chukwugoziem Ihekweaba

RESEARCH EXPERIENCE

Canadian Institute for Cybersecurity, UNB, Fredericton, CA **2025**
Visiting Research Student

- Conducted research on “Federated Device Fingerprinting and Identification for Enhanced Intrusion Detection in Evolving IoT Systems”.
- Developed and published several articles and conference paper with Q1 journals including Elsevier Internet of Things, Expert Systems with Application.
- Received the Best Poster Award at the PST2025 Conference held in Fredericton, Canada.

Universidade Federal de Lavras, Minas Gerais, Brazil **2022**
Postgraduate Research

- Designed and implemented an original research: “An Intelligent Network Planning Tool for 5G and Beyond Networks using Unmanned Arial Vehicle Base Stations”.
- Designed and executed a novel research: “Multi-phase Intrusion Detection System for Computer Networks Based on Deep Learning Algorithms”.
- Participated in a project on Detection of COVID-19 using Deep Learning Algorithms.

Michael Okpara University of Agriculture, Umudike, AB, NG **2016**
Undergraduate Research

- Designed and executed original research: “Computer-based Mobile Surveillance Robot using Arduino and C# Language”.

PROFESSIONAL MEMBERSHIP

Member, Institute of Electrical and Electronic Engineering, IEEE
Member, International Association of Engineers, IAENG
Member, Nigerian Society of Engineers, NSE
Member, Association of Cybersecurity Professionals, ACSP
Registered Engineer, Council for the Regulation of Engineering in Nigeria, COREN
Member, National Institute of Professional Engineers and Scientists, NIPES

ACADEMIC PUBLICATIONS

Ogobuchi Daniel Okey, Sajjad Dadkhah, Demóstenes Zegarra Rodríguez, João Henrique Kleinschmidt, “Explainable Resource-Aware IoT security model via knowledge distillation and adaptive loss function optimization,” **Expert Systems with Applications,** Vol. 302, 130460, 2026, **doi: 10.1016/j.eswa.2025.130460.**

Ogobuchi Daniel Okey, Sajjad Dadkhah, Heather Molyneaux, Demóstenes Zegarra Rodríguez, João Henrique Kleinschmidt, “iPASecIoT: An intelligent pipeline for automatic and adaptive feature extraction for secure IoT device identification and intrusion detection,” **Internet of Things,** Vol. 34,101802, 2025, **doi.: 10.1016/j.iot.2025.101802.**

Ogobuchi Daniel Okey, Rodríguez Demóstenes Zegarra, Guimarães Frederico Gadelha, Kleinschmidt João Henrique, "CAFiKS: Communication-Aware Federated IDS With Knowledge Sharing for Secure IoT Connectivity," **IEEE Access**, Vol. 13, 2025, pp. 140694-140708, **doi.: 10.1109/ACCESS.2025.3596374**

Ogobuchi Daniel Okey, Rodriguez, Demóstenes Zegarra, Kleinschmidt, João Henrique, "Enhancing IoT Intrusion Detection with Federated Learning-Based CNN-GRU and LSTM-GRU Ensembles," in **2024 19th International Symposium on Wireless Communication Systems (ISWCS),** 2024, Pp. 1-6, **doi.: 10.1109/ISWCS61526.2024.10639159**

Ogobuchi Daniel Okey, Ekikere Umoren Udo, Renata Lopes Rosa, Demóstenes Zegarra Rodríguez, João Henrique Kleinschmidt, “Investigating ChatGPT and cybersecurity: A perspective on topic modeling and sentiment analysis,” in **Computers & Security,** Vol. 135, 103476, 2023, **doi: 10.1016/j.cose.2023.103476.**

O. D. Okey, D. C. Melgarejo, M. Saadi, R. L. Rosa, J. H. Kleinschmidt and D. Z. Rodríguez, "Transfer Learning Approach to IDS on Cloud IoT Devices Using Optimized CNN," in **IEEE Access**, vol. 11, pp. 1023-1038, 2023, **doi: 10.1109/ACCESS.2022.3233775.**

Ogobuchi Okey, D.; Sarah Maidin, S.; Adasme, P.; Lopes Rosa, R.; Saadi, M.; Carrillo Melgarejo, D.; Zegarra Rodríguez, D. BoostedEnML: Efficient Technique for Detecting Cyberattacks in IoT Systems Using Boosted Ensemble Machine Learning. **Sensors 2022, 22(19), https://doi.org/10.3390/s22197409**

Okey Daniel Ogobuchi, Samuel Terra Vieira, Muhammad Saadi, Renata Lopes Rosa, and Demóstenes Zegarra Rodríguez: An Intelligent Network Planning tool for Location Optimization of Unmanned Aerial Vehicle Base Stations using Geographical Images**. Journal of Electronic Imaging, 31**(6), 061822 (2022), **10.1117/1.JEI.31.6.061822.**

Arthur A. M. Teodoro, Douglas H. Silva, Muhammad Saadi, **Ogobuchi D. Okey**, Renata L. Rosa, Sattam Al Otaibi & Demóstenes Z. Rodríguez: An Analysis of Image Features Extracted by CNNs to Design Classification Models for COVID-19 and Non-COVID-19. **J Sign Process Syst** (2021). https://doi.org/10.1007/s11265-021-01714-7.

Kazaure, J. S., Matthew, U. O., Okafor, N. U., & **Okey, O. D.** (2021). Telecommunication Network Performances and Evaluation of Radio Frequency Electromagnetic Radiation: Health Effects of the RF-EMR GSM Base Stations**. International Journal of Information Communication Technologies and Human Development (IJICTHD),** 13(3), 16-37. **http://doi.org/10.4018/IJICTHD.2021070102.**

PROFESSIONAL EXPERIENCES

Michael Okpara University of Agriculture, Umudike. Abia, NG
Graduate Assistant, Department of Computer Engineering 2019 – present

- Assisting in teaching and researching
- Guiding and advising students in their research projects
- Administering, invigilating and marking of examinations questions
- As an academic advisor, provide the students with all necessary information that will assist in their academic success.

Splashnet Technologies Limited, Lagos, NG
Data Analyst and Project Development Lead 2018 – 2019

- Collected, analyzed and prepared the results of over 5,000 students using the school management solution.
- Generated progress reports for the students and staff of the school.
- Assisted lecturers in the preparation and uploading of students' academic results to the server
- Trained over 3000 staff and students on how to use the educational solution.
- Supervised and ensured the development of new software for primary and secondary schools in Lagos managed by the company.
- Provided technical support to staff when the need arose and ensured the coherency of the information in the database.

Federal College of Education, Osiele Abeokuta, NG
Data and System Analyst 2017 – 2018

- Coordinated the software team to develop highly scalable education management platforms.
- Provided regular backup and update of the systems and solutions to ensure that there is no data loss.
- Assisted the chief system analyst in the installation and configuration of over 2,000 newly procured computer systems.
- Assisted in the administration of Computer-Based Test (CBT) during the semesters and admission processing.

ExWhyZee Technologies Limited, Umuahia Abia, NG
Database and Content Manager 2015 – 2015

- Collected, prepared and uploaded various contents to the company's e-commerce website on an average of 100 contents daily.
- Used highly level communication skills to write attractive product description to attract the attention of potential customers.
- Assisted in the acquisition of products by going to malls and taking photos of products which as prepared and uploaded to the e-commerce platform.

Evergreen Comprehensive College Oriuzor, Ebonyi, NG
Science Teacher 2009 – 2011

- Taught Mathematics, Chemistry and Physics to over 200 students in both junior and senior cadres
- Prepared and evaluated their performances during and after the examinations.

AWARDS AND EXTRA-CURICULRA ACTIVITES

Fundação de Amparo à Pesquisa do Estado de Sao Paulo (FAPESP)

- Grant for research internship abroad (March 01 – August 31, 2025) **Grant No: 2024/12903-0**

Fundação de Amparo à Pesquisa do Estado de Sao Paulo (FAPESP)

- Grant for doctoral research (2023 – 2027) **Grant No: 2023/07184-1**

Coordenação de Aperfeicoamento de Pessoal de Nível Superior (CAPES)

- Sponsorship for doctoral research 2022 – 2023 **Grant No: 001**

Forum for Agricultural Research for Africa and Tertiary Education Trust Fund (FARA/TETFund) Award, 2020 – 2022.

- Received for postgraduate (Master) program tenable in Brazil.

Agbami Medical and Engineering Professional Scholarship (AMEPS) Award, 2012 – 2016.

- Award received during my undergraduate program for a period 4 years which was renewed every year based on academic performance; tenable in Nigeria.

Presbyterian Students' Fellowship, Nigeria. (2011 – 2016)

- Meritorious award of excellent service as a distinguished leader who worked for the progress of the students in both local and national environs. During this period, I served as president up to the national level where I was placed with the responsibility of overseeing the welfare of more than 5,000 students scattered across over 200 universities in Nigeria.

REFERENCES

- Prof. Dr. Demóstenes Zegarra Rodríguez, demostenes.zegarra@ufla.br , +55 35 9116-3556
  Professor of Telecommunication and Multimedia Services
  Department of Computer Science,
  Universidade Federal de Lavras, MG, BR.

- Prof. Dr. João Henrique Kleinschmidt, joao.kleinschmidt@ufabc.edu.br, +55 11 98386-2027
  Professor of Information Security
  Department of Information Engineering
  Universidade Federal do ABC, Santo Andre, Sao BR.

- Engr. Dr. Chinenye Matthew-Emmanuel Ezeh, cm.ezeh@mouau.edu.ng , +234 806 940 9372
  Lecturer
  Department of Computer Engineering
  Michael Okpara University of Agriculture, Umudike, Nigeria.